Lesson 1.5: Deployment and DevOps

What you learn

- **CI/CD pipelines.** Continuous integration runs tests on every change; continuous delivery deploys passing builds. GitHub Actions is a common way to define them.
- **Vercel and Render.** Vercel suits frontends and Next.js; Render runs servers, databases and background workers.
- **Docker.** A container packages the app with its dependencies so it runs the same everywhere. A Dockerfile lists the steps.
- **AWS S3 asset hosting.** S3 stores files such as images. Keep buckets private by default and serve public assets deliberately, often through a CDN.

Tools and resources (official sites, not checked live)

- GitHub Actions
- Vercel
- Render
- Docker
- Amazon S3

ASSIGNMENT

Practice for lesson 1.5

Business problem: Ship the tasks API.

Steps

1. Write a Dockerfile.
2. Add a workflow that runs tests on push.
3. Deploy to Render.
4. Store environment variables in the host, not in the repository.

Expected output: A public URL that returns your tasks and a green workflow run.

SOLUTION (ANSWER KEY)

Secrets live in the platform's environment settings. The workflow fails the build if tests fail, so broken code never deploys.

Check yourself

1. Why keep secrets out of the repo?
   Answer: Anyone with the code could use them.
2. What does CI do?
   Answer: Runs tests automatically on each change.

Lesson 1.5 of the **Development and IT** lessons. See this lesson and its official links on skillnavora.in · All chapters and lessons · Open the full lesson guide. Tools, prices and plans change; confirm on each official site.